

Fig. 2: An actual-size PCB for the testing circuit

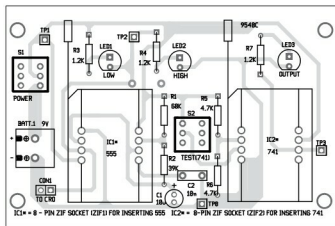


Fig. 3: Component layout for the PCB

Between pin 5 of IC1 and ground a 10nF ceramic capacitor (C2) is used to filter out any noise.

The 741 tester section is wired in a simple comparator mode. Around half of the supply voltage is produced through divider circuit comprising resistors R5 and R6 at inverting input pin 2 of IC2.

For testing 741, close switch S2, if LED3 glows properly the 741 is in good condition.

Construction and testing

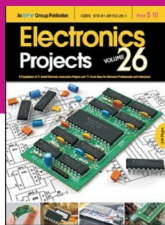
An actual-size, single-side PCB for the 555 and 741 ICs testing circuit is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on a PCB, enclose it in a suitable plastic box. Solder two 8-pin ZIF sockets ZIF1 and ZIF2 on the PCB for 555 timer and 741 op-amp, respectively.

During testing, both the ICs should be inserted in their respective sockets. You can check frequency output of IC1 by connecting an oscilloscope's probes to CON1 in the PCB. In case of any problem, verify that voltages at test points are as per table. ●

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